Define Problem Statements

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	3 Marks

Core Problem Statements & Root Cause Analysis

ID	Problem Statement	Root Cause	Impact on Investigation
PS-01	**Fragmented Evidence Synchronization:** Visual, audio, and video streams analyzed in isolation without a common temporal clock.	Absence of multimodal cross-referencing platforms capable of fusing time-stamped video with speech.	Critical timeline discrepancies, high investigator burnout, and delayed judicial proceedings.
PS-02	**Loss of 3D Spatial Realism:** Flat 2D photographs fail to convey true physical distances and weapon trajectories.	Reliance on static photography and manual tape-measure sketches without depth estimation.	Vulnerability to aggressive defense cross-examination in court regarding line-of-sight.
PS-03	**Language Barriers in Testimonies:** Eyewitnesses in multicultural urban zones speak regional/foreign languages.	Lack of automated, immediate forensic-grade multilingual speech-to-text workflows.	Vital initial golden-hour investigative leads are lost while awaiting official translators.
PS-04	**Inconsistent & Slow Documentation:** Generating case reports requires hours of manual drafting prone to error.	Manual collation of handwritten notes and audio transcripts without cryptographic validation.	Chain-of-custody vulnerabilities and delays in submitting court-admissible dossiers.